LLM4HRS: LLM-Based Spatiotemporal Imputation Model for Highly Sparse Remote Sensing Data

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***Abstract*— Remote sensing data are of considerable significance for monitoring global climate, detecting harmful algae bloom, and so on. However, due to sensor failures, cloud cover, and thick aerosols, the collected remote sensing data for various ocean factors, such as chlorophyll-a (Chl-a) concentration and sea surface temperature (SST), often have a high missing rate, which seriously hinders their applications. Existing data imputation models mostly ignore highly sparse spatial locations or perform badly when the data missing rate is high due to the lack of available information. Large language models (LLMs) possess powerful representation learning capabilities and can effectively capture sequential correlations even with extremely limited information, thereby presenting the promising potential for highly sparse remote sensing data imputation. Therefore, we proposed a novel LLM-based spatiotemporal model for highly sparse remote sensing data imputation, i.e., LLM4HRS. First, LLM4HRS develops an LLM-based bidirectional temporal representation learning module to learn forward and backward temporal dependencies in data sequences and fuses them together to obtain comprehensive temporal representations. Next, LLM4HRS constructs a LLM-based spatial representation learning module to learn spatial correlations with the learned temporal representation. Finally, a spatiotemporal representation fusion and data imputation module is developed to achieve data imputation. Experiments on the SST and Chl-a remote sensing datasets demonstrate that LLM4HRS significantly outperforms existing data imputation models, with its advantages becoming more pronounced as the masking rate increases. Furthermore, when extended to the remote sensing PAR data in a large region, LLM4HRS still achieves the best performance, further validating its broad applicability for remote sensing data imputation. The code of LLM4HRS is publicly available at https://github.com/ssyuwang/LLM4HRS-master.**

***Index Terms*— Data imputation, high data sparsity, large language model (LLM), remote sensing data, spatiotemporal modeling.**

I. Introduction

With the rapid development of remote sensing technology, spatiotemporal remote sensing data collected by remote sensing sensors, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) [1], the Visible Infrared Imaging Radiometer Suite (VIIRS) [2], [3], and the Advanced Very High Resolution Radiometer (AVHRR) [4], have laid the foundation for exploring the ocean with machine learning and artificial intelligence technology. For example, remote sensing data for chlorophyll-a (Chl-a) concentration and sea surface temperature (SST) have been widely used to monitor harmful algal bloom [5], [6], [7], [8], [9], [10], learn ocean ecosystem dynamics [11], [12], and analyze ocean primary productivity [13], [14].

However, in practice, remote sensing data usually have a high missing rate due to sensor failures, cloud cover, thick aerosols, and so on. For example, Fig. 1 illustrates the spatiotemporal distributions of missing rates from Ocean Color Climate Change Initiative (OC-CCI) Chl-a data and MODIS-Aqua SST data in the region of Yangtze River Estuary in 2023. As shown in Fig. 1(a) and (b), the missing rates higher than 90% or even missing for the whole day occur frequently. The average daily missing rates for Chl-a and SST are 66.21% and 64.19%, respectively. According to Fig. 1(c) and (d), the average missing rates for most areas are concentrated between 60% and 70%, and the sparsity issue is particularly serious in the areas near the coastline. In conclusion, remote sensing data are highly sparse in both temporal and spatial dimensions, which seriously hinders various applications highly dependent on remote sensing data. Therefore, it is indispensable to design a spatiotemporal data imputation model for highly sparse remote sensing data to accurately fill in the missing values and provide complete remote sensing data of high quality.

Deep learning models are suitable for remote sensing data imputation due to their excellent ability to learn complex nonlinear correlations and underlying data distributions from large-scale data [15], [16], [17], [18]. However, with increasing

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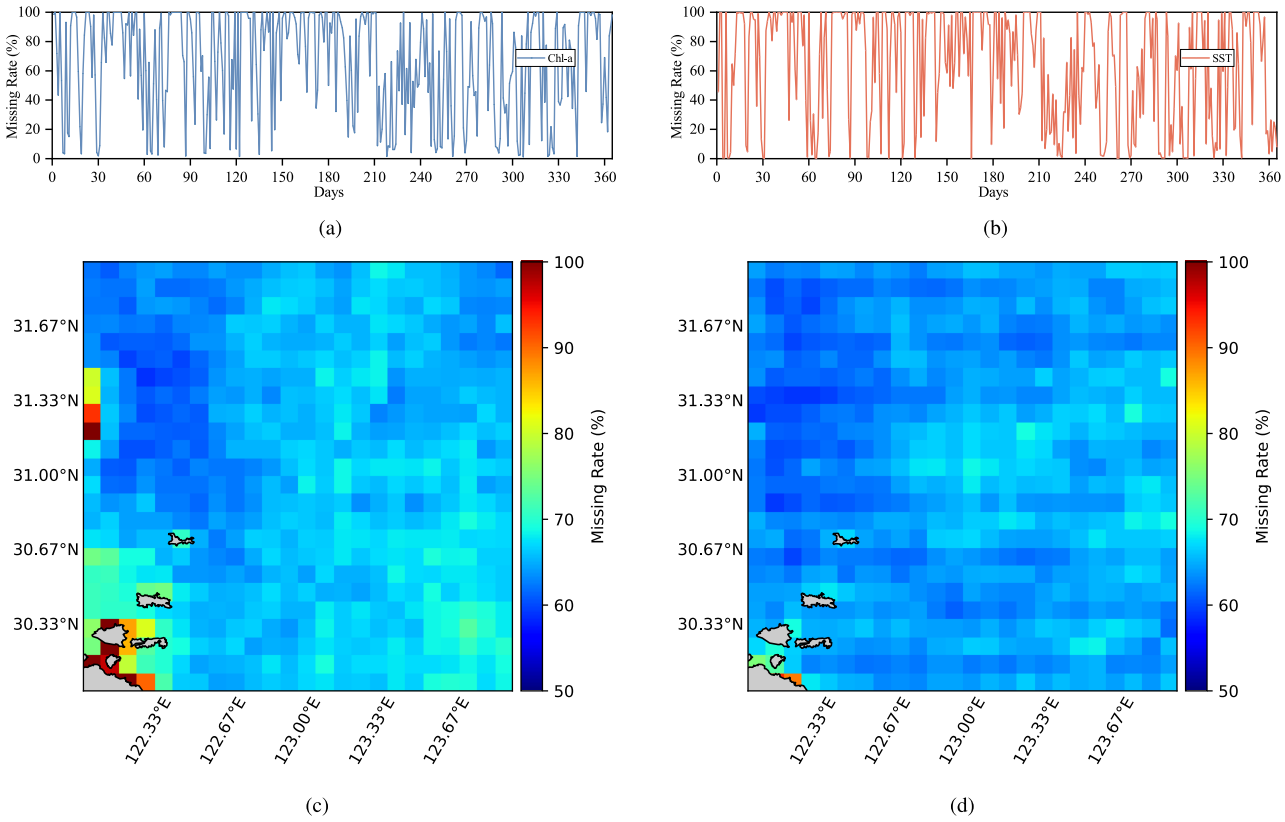


Fig. 1. Temporal and spatial variations of missing rates of OC-CCI Chl-a and MODIS-Aqua SST data over the region of Yangtze River Estuary in 2023, where gray regions correspond to land. (a) Variation of daily missing rates of OC-CCI Chl-a data in 2023. (b) Variation of daily missing rates of MODIS-Aqua SST data in 2023. (c) Spatial variation of missing rates of OC-CCI Chl-a data. (d) Spatial variation of missing rates of MODIS-Aqua SST data.

masking rate, the data imputation performance of deep learning models decreases drastically due to the limited available information about the masked records. Recently, the powerful representation learning ability demonstrated by the large language model (LLM) in the field of natural language processing (NLP) has attracted widespread attention from scholars in the field of time series analysis [19], [20], [21], [22]. For example, Zhou et al. [21] proposed the LLM-based time series analysis model (OFA) that applies LLM to time series analysis tasks, including data imputation. OFA has shown surprisingly superior results compared to state-of-the-art models using the same backbone, which is mainly attributed to its powerful feature learning capability. The advantage of LLM makes it suitable for learning data distributions in the absence of a large amount of ground truth and motivates its use in achieving highly sparse remote sensing data imputation. However, applying LLM models directly to highly sparse remote sensing data cannot fully utilize their feature representation learning capability, mainly due to the following two challenges.

On the one hand, remote sensing data typically exhibit bidirectional temporal dependence as both historical and future time series are already known, and the targets to be imputed often depend on these two time series. Moreover, the dependencies on historical and future data usually change dynamically over time according to certain temporal patterns. Fig. 2(a) shows an illustration of bidirectional temporal dependence for a specific time step without observation value. In this case, the values for the target time step t and the adjacent historical time steps $t-2$ and $t-1$ are missing. Relying solely on data from more distant historical time steps $t-5$, $t-4$, and $t-3$ to fill in the value for time step t could lead to significant bias. However, data from future time steps $t+1$, $t+2$, and $t+3$ exhibit a similar temporal pattern to that of time step t. Incorporating the values of these future time steps can greatly improve the imputation accuracy for time step t. Therefore, the bidirectional temporal dependence in remote sensing data is important for data imputation, but it is challenging for current LLM-based sequential models to capture such dependencies effectively.

On the other hand, remote sensing data exhibit the characteristics of high spatial correlations. We visualize the Chl-a of four locations in the study area of the Yangtze River Estuary from 2019 to 2023 [see Fig. 2(b)]. It is obvious that the locations in the same regions, e.g., location pairs (A, B) and (C, D), have similar data distributions, but locations in different regions have diverse Chl-a data distributions, e.g., location pairs (A, C) and (B, D). Obtaining relevant information from spatial neighbors is crucial for highly sparse remote sensing data imputation. However, current LLM-based models are channel-independent, which means that they do not consider spatial dependencies. This limitation hinders their applications in highly sparse remote sensing data imputation.

In this work, to address the issue of highly sparse, we propose an LLM-based spatiotemporal model (LLM4HRS) to effectively learn and fuse the temporal and spatial data representations for accurate highly sparse remote sensing data

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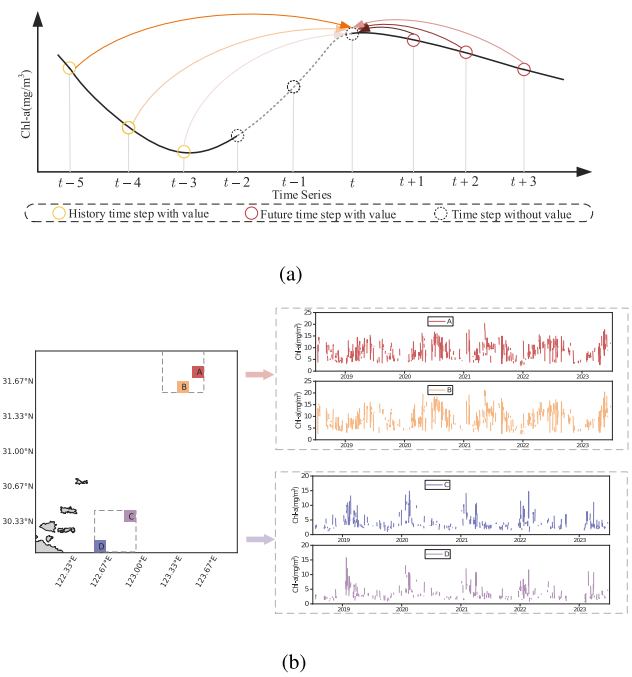


Fig. 2. Illustration of bitemporal dependence and spatial correlation. (a) Illustration bidirectional temporal dependence, where time step t is the target missing point to be filled and curved lines with arrows represent dependence weights, with darker colors indicating greater weights. (b) Locations in the same regions tend to have similar Chl-a data distributions, while different regions have diverse Chl-a data distributions.

imputation. Specifically, our main contributions include the following.

1) We proposed the novel LLM-based spatiotemporal model LLM4HRS for highly sparse remote sensing data imputation.
2) We developed an LLM-based bidirectional temporal representation learning module to learn both forward and backward temporal data representations, which are then fused to achieve comprehensive learning of the temporal representations of remote sensing data.
3) We designed an LLM-based spatial representation learning module that inverts the temporal and spatial attributes of the learned bidirectional temporal representations and feeds them into a frozen pretrained LLM to learn spatial representations of remote sensing data.
4) We developed a spatiotemporal representation fusion and data imputation module that fuses the learned bidirectional temporal and spatial representations to generate data, which is then used to replace missing values in highly sparse remote sensing data, thereby achieving data imputation.
5) We conducted extensive experiments and ablation studies on four real-world remote sensing datasets to evaluate the performance of LLM4HRS model and justify its design. The results indicate that the LLM4HRS model outperforms the latest data imputation models by a large margin.

The rest of this work is organized as follows. Section II reviews the related work on remote sensing data imputation. Section III describes the datasets and study areas. Section IV details the proposed LLM4HRS model for highly sparse remote sensing data imputation. Then, we present the experiments and results in Section V and conduct the discussion in Section VI. Finally, we draw the conclusion in Section VII.

II. Related Works

Remote sensing data imputation aims to utilize the limited available spatiotemporal data to reconstruct all the missing values. The existing models for remote sensing data imputation can be roughly divided into *statistical models* and *data-driven models*.

A. Statistical Models

The basic idea of statistical models is to model the linear or nonlinear relationships in remote sensing data under certain statistical assumptions to impute missing values. The optimal interpolation (OI) model [23], [24], data interpolating empirical orthogonal functions (DINEOF) [25], [26], [27], [28], and Kriging [29] are commonly used statistical models for remote sensing data imputation. For example, SST reanalysis data products often utilize the OI model to impute missing values [23], [24]. Wang et al. [26] use the DINEOF to complete monthly SST and Chl-a data. Ye et al. [27] also utilize DINEOF to preprocess Chl-a data for filling in the missing values in both five-day and monthly datasets. Statistical models make certain assumptions about the distributions and correlations of remote sensing data and usually can only handle simple nonlinear relationships. Such limitations result in the models being unable to effectively capture complex features of remote sensing data for accurate data imputation.

B. Data-Driven Models

With the rapid development of machine learning techniques, the data-driven models for remote sensing data imputation are of increasing popularity in recent years [30], [31]. Existing data-driven models for remote sensing data imputation can be further categorized into *traditional machine learning models* and *deep learning models*.

1) Traditional Machine Learning Models: Traditional machine learning models learn hidden patterns in remote sensing data directly to fill in the missing values. Representative models include a random forest (RF), a self-organizing map (SOM), an artificial neural network (ANN), an extremely randomized tree (ExtraTree), an extreme gradient boosting (Xgboost), and a support vector regression (SVR) [32], [33], [34], [35], [36]. For example, Li et al. [33] employ traditional machine learning techniques such as RF for Chl-a data imputation in the South China Sea region. Sunder et al. [32] utilize ANN, SVR, and RF for MODIS-Aqua SST data imputation. Park et al. [34] employ ensemble learning models, e.g., RF and ExtraTree, for sparse Chl-a data imputation using cloud-free microwave and reanalysis data. In general, traditional machine learning models cannot effectively model the complex spatiotemporal dependencies and dynamic regularity in remote sensing data.

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2) Deep Learning Models: Deep learning models can effectively learn the complex representations of remote sensing data, making them suitable for remote sensing data imputation. For example, Barth proposed the data-interpolating convolutional autoencoder (DINCAE) model [37] for filling the missing values in AVHRR SST data.

DINCAE is also widely used for the imputation of Chl-a data [38]. Barth et al. [39] extended DINCAE to handle multivariate data, including SST, Chl-a, and wind speed, for reconstructing MODIS-Terra SST and altimetry observations. Chen et al. [40] combined Fourier transform convolution with LSTM for monthly Chl-a data imputation. Generative adversarial networks (GANs), known for their strength in reconstructing the original data distribution, are widely applied in remote sensing data imputation. Shibata et al. [41] were the first to employ GANs for the reconstruction of SST images. In our previous research [42], we incorporated graph attention networks (GATs) into a GAN framework to capture the spatiotemporal dependencies in remote sensing data, enabling accurate imputation of Chl-a and SST data. However, the above studies cannot work well on highly sparse data imputation due to the issue of limited available information. To address this issue, Wang et al. [43] integrated external SST and photosynthetically active radiation (PAR) data to enrich Chl-a feature representations, thereby improving the accuracy of highly sparse Chl-a data imputation. Ye et al. [44] combined a statistical model with the attention-based SwinUnet network to reconstruct highly sparse Chl-a data. Hirahara et al. [45] presented an anomaly inpainting network based on GAN for SST reconstruction.

Recently, most high-performing models for time series data imputation are based on attention mechanisms, multiperiodic analysis, and a temporal convolutional network (TCN). Exemplary attention-based models include SAITS [18], iTransformer [46], and OFA [21]. These models learn the data distributions by dynamically capturing temporal dependencies. In addition, Crossformer [47], known for its success in time series forecasting, has also been applied to data imputation. TimesNet [48] is another notable data imputation model based on multiperiod analysis and leverages fast Fourier transform (FFT) to extract features of different periods to uncover complex temporal dependencies. The ModernTCN model [49] captures temporal and spatial dependencies solely through TCN and outperforms most attention-based models for time series data imputation. These deep learning-based imputation models can be directly applied to highly sparse remote sensing data imputation and usually outperform statistical models and traditional machine learning models.

C. Discussion

Though deep learning-based models achieve state-of-the-art performance in remote sensing data imputation, they are still facing some challenges. For example, they have not fully utilized the spatiotemporal dependencies of limited available data to reconstruct highly sparse remote sensing data. Meanwhile, the powerful representation learning capability of LLM has attracted an increasing number of researchers to use it for time series analysis [19], [21], [50], [51]. Therefore, we proposed the novel LLM-based spatiotemporal model LLM4HRS for highly sparse remote sensing data imputation. Specifically, considering the strong spatiotemporal dependence in ocean remote sensing data, we develop LLM-based bidirectional temporal representation learning and LLM-based spatial representation learning to fully leverage the representation learning capability of LLM.

III. Data and Study Areas

A. Datasets

We take Chl-a and SST as examples to demonstrate the idea of our model and evaluate its effectiveness.

1) Chl-a Data: The Chl-a data are from OC-CCI 6.0 product,[1] which fuses the satellite data from MODIS, VIIRS, Sea-Viewing Wide Field-of-View Sensor (SeaWiFS), and Medium Resolution Imaging Spectrometer (MERIS) sensors to produce daily Chl-a records. The spatial resolution is 4 km ($1/24^\circ \times 1/24^\circ$), and the time range is from September 1997 to December 2023.

2) SST Data: The SST data are from AVHRR[2] and collected twice every day. The spatial resolution is also 4 km ($1/24^\circ \times 1/24^\circ$), and the time range is from September 1981 to December 2023. We choose day data as the study dataset and only retain the data with a quality level of 4 or higher to guarantee the data quality.

B. Study Areas

To unify the time range of the three datasets of Chl-a and SST, we study the time series range from January 1, 2000, to December 31, 2023. Two study areas, i.e., East China Sea and Yangtze River Estuary, are selected, and their details are elaborated on as follows.

1) EAST: The study area of East China Sea covers the spatial region of [128.00°E–130.00°E, 24.00°N–26.00°N], with a total of 2304 (48×48) grid regions, and the size of each grid region is 4×4 km. We denote Chl-a and SST as EAST-CHA and EAST-SST, respectively, in this area. The data missing rates of EAST-CHA and EAST-SST are 68.57% and 89.37%, respectively. Fig. 3 shows the spatial distribution of EAST-CHA and EAST-SST on August 17, 2023.

2) YRE: The study area of Yangtze River Estuary covers the spatial region of [122.00°E–124.00°E, 30.00°N–32.00°N], with a total of 2304 (48×48) grid regions, and the size of each grid region is 4×4 km. We denote Chl-a and SST as YRE-CHA and YRE-SST, respectively, in this area. The data missing rates of YRE-CHA and YRE-SST in this area are 80.22% and 90.93%, respectively. Fig. 4 shows the spatial distributions of YRE-CHA and YRE-SST on August 4, 2023. Compared with the study area of EAST, the Chl-a and SST data in this area exhibit significant fluctuations since YRE is closer to the coastline.

Note that the LLM GPT-2 used in this study requires approximately 117-MB parameters, resulting in quite a high

[1] https://esa-oceancolour-cci.org

[2] https://www.ncei.noaa.gov/data/oceans/pathfinder/Version5.3/

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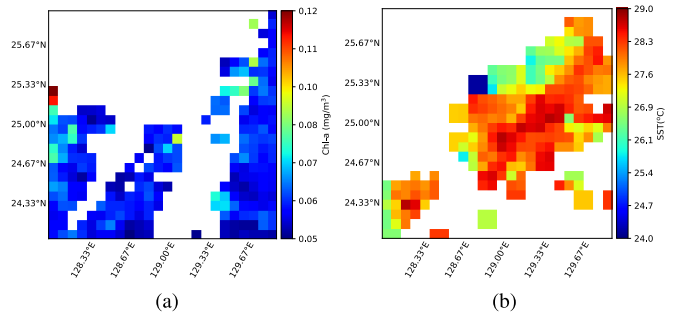


Fig. 3. Spatial distributions of Chl-a and SST in the study area of East China Sea on August 17, 2023, where the white areas indicate no data. (a) EAST-CHA. (b) EAST-SST.

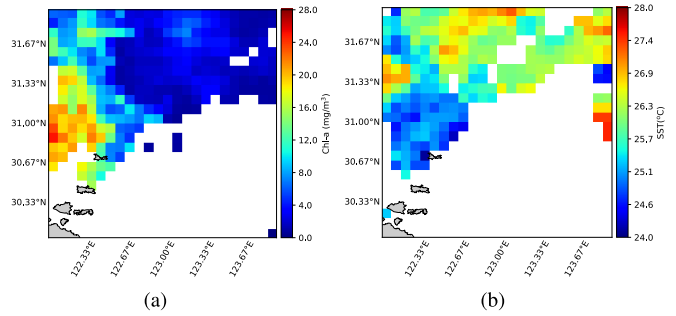


Fig. 4. Spatial distributions of Chl-a and SST in the study area of Yangtze River Estuary on August 4, 2023, where the white areas indicate no data and gray areas represent land. (a) YRE-CHA. (b) YRE-SST.

computation cost. To alleviate this issue, we reduce the number of grid regions in each study area to 576 (24 × 24) via a downsampling operation, with each region of size 8 × 8 km.

IV. Method

A. Problem Definition

Highly sparse remote sensing data imputation is usually formulated as a spatiotemporal imputation problem, i.e., learning the spatiotemporal distribution of remote sensing data based on the limited observed remote sensing data to reconstruct the missing data. Specifically, remote sensing data are denoted as $\boldsymbol{\mathcal{X}}=\{X_1, X_2, \ldots, X_T\}\in\mathbb{R}^{T\times N}$, where T is the number of time steps, N is the number of grid regions, and $X_t=\{x_{t,1}, x_{t,2}, \ldots, x_{t,N}\} \in \mathbb{R}^{1\times N}$ $(1 \le t \le T)$ records the remote sensing data observations for N locations at time step t. The remote sensing data matrix $\boldsymbol{\mathcal{X}}$ is highly sparse, and we introduce a mask matrix $\boldsymbol{\mathcal{M}} \in \mathbb{R}^{T\times N}$ to indicate the status of values in $\boldsymbol{\mathcal{X}}$. The items in $\boldsymbol{\mathcal{M}}$ are determined as follows:

$$m_{t,i} = \begin{cases} 1, & \text{if } x_{t,i} \text{ is observed} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For the remote sensing data imputation problem, based on the data matrix $\boldsymbol{\mathcal{X}}$ and the corresponding mask matrix $\boldsymbol{\mathcal{M}}$, we aim to find a function $\mathcal{F}$ to fill the missing values in $\boldsymbol{\mathcal{X}}$ to obtain the complete data matrix $\widetilde{\boldsymbol{\mathcal{X}}}$, i.e.,

$$\widetilde{\boldsymbol{\mathcal{X}}} = \mathcal{F}_\theta(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{M}}) \tag{2}$$

where θ denotes the learnable parameters in $\mathcal{F}$.

B. Model Overview

Fig. 5 shows the architecture of LLM4HRS model, which consists of the LLM-based bidirectional temporal representation learning module, the LLM-based spatial representation learning module, and the spatiotemporal representation fusion and data imputation module. Specifically, the LLM-based bidirectional temporal representation learning uses two frozen pretrained LLM blocks to separately learn forward and backward temporal dependencies in remote sensing data. Subsequently, the learned bidirectional temporal representation $\widehat{\boldsymbol{\mathcal{X}}}_T \in \mathbb{R}^{T\times N}$ is inverted to perform LLM-based spatial representation learning to learn spatial correlations $\widehat{\boldsymbol{\mathcal{X}}}_S \in \mathbb{R}^{T\times N}$. Finally, the temporal and spatial representations are fused using a linear layer to obtain the final generated data $\widehat{\boldsymbol{\mathcal{X}}}$, which is subsequently utilized for replacing the missing values in sparse remote sensing data $\boldsymbol{\mathcal{X}}$ to achieve data imputation.

C. LLM-Based Bidirectional Temporal Representation Learning

Due to the bidirectional temporal dependence characteristic of sparse remote sensing data, solely considering historical data information during the data imputation process fails to fully learn the temporal patterns and may result in unsatisfactory data imputation accuracy. To address this issue, we introduce an LLM-based bidirectional temporal representation learning module, which applies pretrained LLMs to learn the forward and backward temporal representations separately and then fuses them together to obtain comprehensive temporal representation. The LLM-based bidirectional temporal representation learning module consists of three components, i.e., forward temporal representation learning, backward temporal representation learning, and bidirectional temporal representation fusion.

1) Forward Temporal Representation Learning: We utilize forward time series, i.e., time steps from 1 to T, to learn the temporal representation. To this end, we first design the forward temporal embedding $\mathbf{E}_T^{fwd}$ for highly sparse remote sensing data $\boldsymbol{\mathcal{X}}$ to fit the dimension of the pretrained LLM, i.e.,

$$\mathbf{E}_T^{fwd} = \text{Linear}(\boldsymbol{\mathcal{X}}||\boldsymbol{\mathcal{M}}) \tag{3}$$

where Linear$(\cdot)$ is the mapping of $\mathbb{R}^{T\times 2N} \to \mathbb{R}^{T\times d_{\text{model}}}$ and d_{model} is the input dimension of the LLM model. To better learn the forward temporal representation of $\boldsymbol{\mathcal{X}}$, we concatenate $\boldsymbol{\mathcal{X}}$ with the corresponding mask matrix $\boldsymbol{\mathcal{M}}$ as the input of forward temporal representation learning.

We choose the GPT-2 model [52] as the pretrained LLM model. GPT-2 is a pretrained LLM based on the transformer architecture and is shown in Fig. 6. Inspired by Lu et al. [53], the multihead self-attention layers and feedforward layers are frozen during the fine-tuning. Since GPT-2 is an LLM model designed for NLP, fine-tuning positional embedding and layer normalization is conducted to adapt GPT-2 to the remote sensing data imputation task.

1) *Fine-Tuning Positional Embedding and Layer Normalization Blocks:* We use the positional embedding and layer normalization modules consistent with OFA [21].

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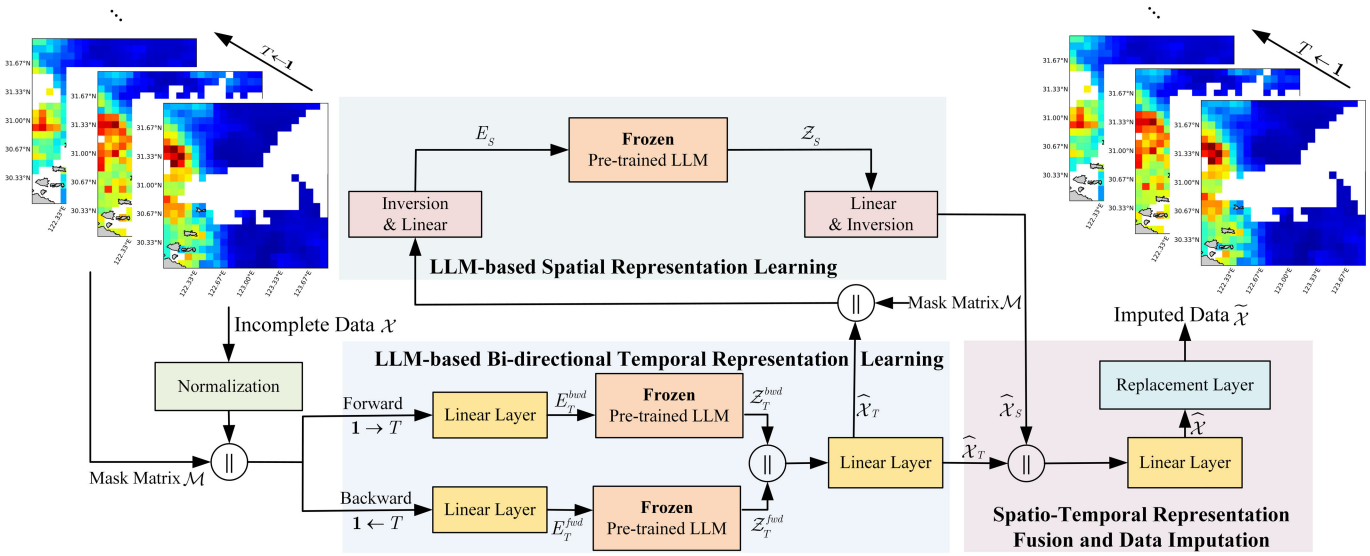


Fig. 5. Architecture of LLM4HRS model that consists of the LLM-based bidirectional temporal representation learning module, the LLM-based spatial representation learning module, and the spatiotemporal representation fusion and data imputation module, where || represents the concatenation operation.

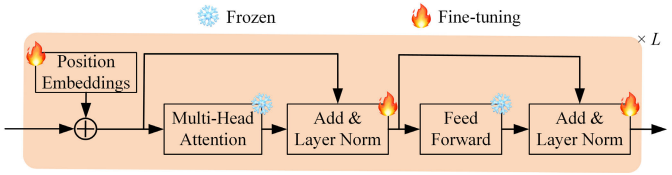


Fig. 6. Framework of the GPT-2 model.

Specifically, the sin and cos functions are used to encode the temporal position in the remote sensing data matrix $\boldsymbol{\mathcal{X}}$, i.e.,

$$\mathrm{PE}^{fwd}_{(\mathrm{pos},2i)} = \sin\left(\mathrm{pos}/10000^{2i/d_{\mathrm{model}}}\right) \tag{4}$$

$$\mathrm{PE}^{fwd}_{(\mathrm{pos},2i+1)} = \cos\left(\mathrm{pos}/10000^{2i/d_{\mathrm{model}}}\right) \tag{5}$$

where pos is the time step position and i is the dimension. For the forward time series, we have pos $= 1, 2, \ldots, T$.

2) *Freezing Multihead Attention and Feedforward Network (FFN) Blocks:* The multihead attention is defined as follows:

$$\mathrm{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{6}$$

$$\mathrm{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = W^{O}(\mathrm{head}_1||\mathrm{head}_2||\ldots||\mathrm{head}_h) \tag{7}$$

where $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are different embeddings of the input data; $\mathrm{head}_i = \mathrm{Attention}(\boldsymbol{Q}W_i^Q, \boldsymbol{K}W_i^K, \boldsymbol{V}W_i^V)$; $W_i^Q \in \mathbb{R}^{d_{\mathrm{model}}\times d_k}$, $W_i^K \in \mathbb{R}^{d_{\mathrm{model}}\times d_k}$, $W_i^V \in \mathbb{R}^{d_{\mathrm{model}}\times d_v}$, and and $W^O \in \mathbb{R}^{hd_v\times d_{\mathrm{model}}}$ are the learnable parameters; and h is the layer of attention.

The FFN has two linear transformation layers with an ReLU activation in the middle, i.e.,

$$\mathrm{FFN}(x) = \mathrm{ReLU}(xW_1 + b_1)W + b \tag{8}$$

where W_1, W, b_1, and b are the learnable parameters.

Finally, the forward temporal representation $\boldsymbol{\mathcal{Z}}_T^{fwd} \in \mathbb{R}^{T\times d_{\mathrm{model}}}$ learned by frozen pretrained LLM can be described as follows:

$$\boldsymbol{\mathcal{Z}}_T^{fwd} = \mathrm{FFN}\left(\mathrm{MultiHead}\left(\mathbf{E}_T^{fwd} + \mathrm{PE}_T^{fwd}\right)\right) \tag{9}$$

where PE_T^{fwd} is the positional embedding for forward time series.

2) Backward Temporal Representation Learning: Similarly, we use the same embedding and representation learning models to obtain the backward temporal representation. The main difference is that the time step order of the remote sensing data is reversed, i.e., changing the time steps of the highly sparse remote sensing data $\boldsymbol{\mathcal{X}}$ and the mask matrix $\boldsymbol{\mathcal{M}}$ from $1 \rightarrow T$ to $1 \leftarrow T$. In this way, we can obtain the backward temporal representation $\mathbf{Z}_T^{bwd} \in \mathbb{R}^{T\times d_{\mathrm{model}}}$ as well.

3) Bidirectional Temporal Representation Fusion: To obtain bidirectional temporal representation $\widehat{\mathcal{X}}_T$, we design a fusion layer to combine the forward temporal representation $\mathcal{Z}_T^{fwd}$ and backward temporal representation $\mathcal{Z}_T^{bwd}$, i.e.,

$$\widehat{\boldsymbol{\mathcal{X}}}_T = \mathrm{Linear}\left(\boldsymbol{\mathcal{Z}}_T^{fwd}||\boldsymbol{\mathcal{Z}}_T^{bwd}\right) \tag{10}$$

where $\mathrm{Linear}(\cdot)$ achieves the mapping of $\mathbb{R}^{T\times 2d_{\mathrm{model}}} \rightarrow \mathbb{R}^{T\times N}$.

D. LLM-Based Spatial Representation Learning

To learn the spatial dependencies in highly sparse remote sensing data, we introduce an LLM-based spatial representation learning module, which consists of spatial embedding and spatial representation learning.

1) Spatial Embedding: The spatial representation learning module focuses on spatial-centric representation learning. To this end, we perform an inversion operation on the bidirectional temporal representation $\widehat{\boldsymbol{\mathcal{X}}}_T$, i.e., the rows (temporal) and columns (spatial) are swapped. The inverted results are

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then fed into a linear layer to obtain the spatial embedding, i.e.,

$$\mathbf{E}_S = \text{Linear}\Big(\text{Inversion}\Big(\widehat{\boldsymbol{\mathcal{X}}}_T\Big)||\text{Inversion}(\boldsymbol{\mathcal{M}})\Big) \tag{11}$$

where Inversion$(\cdot)$ achieves the mapping of $\mathbb{R}^{T\times N} \rightarrow \mathbb{R}^{N\times T}$ and Linear$(\cdot)$ achieves the mapping of $\mathbb{R}^{N\times 2T} \rightarrow \mathbb{R}^{N\times d_{\text{model}}}$. To better learn the spatial dependencies of highly sparse remote sensing data, we concatenate the bidirectional temporal representation $\widehat{\boldsymbol{\mathcal{X}}}_T$ with the corresponding mask matrix $\boldsymbol{\mathcal{M}}$ as the input of LLM-based spatial representation learning module.

2) Spatial Representation Learning: This module has a similar structure to the frozen pretrained LLM module of the bidirectional temporal representation learning, but its inputs are different, i.e.,

$$\boldsymbol{\mathcal{Z}}_S = \text{FFN}(\text{MultiHead}(\mathbf{E}_S + \text{PE}_S)) \tag{12}$$

where PE_S is positional encoding for spatial embedding and $\boldsymbol{\mathcal{Z}}_S \in \mathbb{R}^{N\times d_{\text{model}}}$ is the spatial representation.

Finally, we use a linear layer to generate the spatial representation. To ensure consistency with the dimensions of the bidirectional temporal representation, we perform an inversion operation on the spatial representation again to obtain the final spatial representation $\widehat{\boldsymbol{\mathcal{X}}}_S$, i.e.,

$$\widehat{\boldsymbol{\mathcal{X}}}_S = \text{Inversion}(\text{Linear}(\boldsymbol{\mathcal{Z}}_S)) \tag{13}$$

where Linear$(\cdot)$ is the mapping of $\mathbb{R}^{N\times d_{\text{model}}} \rightarrow \mathbb{R}^{N\times T}$ and Inversion$(\cdot)$ achieves the mapping of $\mathbb{R}^{N\times T} \rightarrow \mathbb{R}^{T\times N}$.

E. Spatiotemporal Representation Fusion and Data Imputation

The generated data $\widehat{\boldsymbol{\mathcal{X}}}$ are then obtained using a linear layer that aggregates representations extracted by the bidirectional temporal representation $\widehat{\boldsymbol{\mathcal{X}}}_T$ and spatial representation $\widehat{\boldsymbol{\mathcal{X}}}_S$

$$\widehat{\boldsymbol{\mathcal{X}}} = \text{Linear}\Big(\widehat{\boldsymbol{\mathcal{X}}}_T||\widehat{\boldsymbol{\mathcal{X}}}_S\Big) \tag{14}$$

where Linear$(\cdot) : \mathbb{R}^{T\times 2N} \rightarrow \mathbb{R}^{T\times N}$.

Finally, we fill the missing values in the highly sparse remote sensing data matrix $\boldsymbol{\mathcal{X}}$ with the generated data matrix $\widehat{\boldsymbol{\mathcal{X}}}$ to obtain the completed data $\widetilde{\boldsymbol{\mathcal{X}}}$, i.e.,

$$\widetilde{\boldsymbol{\mathcal{X}}} = \boldsymbol{\mathcal{X}}\boldsymbol{\mathcal{M}} + (\mathbf{1} - \boldsymbol{\mathcal{M}})\widehat{\boldsymbol{\mathcal{X}}} \tag{15}$$

where $\mathbf{1}$ is an all-ones matrix with the same dimension as the mask matrix $\boldsymbol{\mathcal{M}}$.

V. Experiments and Results

We conduct the experimental evaluation on the highly sparse remote sensing data for Chl-a and SST in the two selected study areas. To validate the superiority of LLM4HRS model, we compare it with 11 state-of-the-art imputation models, including attention-based models (Transformer, Crossformer, SAITS, OFA, and iTrtansformer), the recurrent neural network-based model (GRUD), the empirical orthogonal function-based model (DINEOF), the convolutional neural network-based model (DINCAE) multiperiod-based model (TimesNet), the GAN-based model (S2-STI), and the TCN-based model (ModernTCN). In addition, we evaluate the effectiveness of the model components via ablation studies. In all results, LLM4HRS achieves state-of-the-art data imputation performance.

A. Baseline Models

To evaluate the effectiveness of LLM4HRS model, we compare it with multiple baseline models, including the following.

1) *DINEOF (Beckers and Rixen [28]):* An empirical orthogonal function-based model that utilizes singular value decomposition (SVD) to extract spatiotemporal features and iteratively estimates missing values using known data.
2) *Transformer (Vaswani et al. [17]):* An attention-based model that concurrently collects information from various positions in the sequence using a multihead self-attention mechanism.
3) *GRUD (Che et al. [54]):* A recurrent neural network-based model for time series imputation incorporates a decay mechanism to dynamically adjust the focus on historical information.
4) *DINCAE (Barth et al. [37]):* A convolutional neural network-based model that utilizes convolution to extract local spatial correlations for data imputation.
5) *Crossformer (Zhang and Yan [47]):* A cross-spatial dependence patch time series transformer that introduces full connection among locations to capture cross-spatial dependencies.
6) *TimesNet (Wu et al. [48]):* A multiperiod-based time series analysis model that captures complex temporal features by transforming 1-D time series into a 2-D space tensor.
7) *SAITS (Du et al. [18]):* A self-attention-based data imputation model that uses two diagonally masked self-attention blocks to capture temporal dependencies and introduces a joint optimization model for both imputation and reconstruction tasks to improve the data imputation performance.
8) *OFA (Zhou et al. [21]):* An LLM-based model that is tailored for time series analysis including series data imputation and utilizes GPT-2 to learn the representations of time series data.
9) *ModerTCN (Lou and Wang [49]):* A TCN-based time series imputation model that captures both intravariable dependencies and intervariable dependencies to achieve data imputation.
10) *iTransformer (Liu et al. [46]):* An attention-based model that captures location correlations by inverting the transformer to treat each time sequence as a token.
11) *S2-STI (Wang et al. [43]):* A GAN-based model for highly sparse remote sensing data imputation using auxiliary data to enrich the information of remote sensing data.

B. Experimental Setup

1) Experiment Environment: All data imputation models are implemented using Python>=3.8 and PyTorch

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TABLE I

RESULTS OF LLM4HRS AND BASELINE DATA IMPUTATION MODELS ON THE TWO CHL-A DATASETS, I.E., EAST-CHA AND YRE-CHA, WHILE VARYING THE MASKING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND IMPROVEMENT REPRESENTS THE IMPROVEMENT RATIO OF LLM4HRS AGAINST THE SECOND-BEST RESULTS

Dataset	Model	0.2 RMSE	0.2 MAE	0.2 MAPE	0.2 R^2	0.2 SSIM	0.5 RMSE	0.5 MAE	0.5 MAPE	0.5 R^2	0.5 SSIM
EAST-CHA	DINEOF	0.0311	0.0191	23.95%	0.4274	0.6369	0.0478	0.0361	45.11%	-0.3446	0.4468
	Transformer	0.0180	0.0119	15.28%	0.8004	0.6667	0.0193	0.0128	16.31%	0.7710	0.6398
	GRUD	0.0223	0.0142	18.15%	0.6850	0.6234	0.0230	0.0147	18.82%	0.6740	0.6159
	DINCAE	0.0210	0.0161	18.70%	0.7899	0.7985	0.0225	0.0179	22.85%	0.7370	0.6652
	Crossformer	0.0227	0.0154	20.44%	0.6527	0.6096	0.0242	0.0164	21.15%	0.6296	0.5884
	TimesNet	0.0268	0.0194	24.84%	0.5677	0.5829	0.0318	0.0241	30.76%	0.3748	0.5608
	SAITS	0.0189	0.0126	16.16%	0.7877	0.6482	0.0193	0.0129	16.40%	0.7663	0.6644
	OFA	0.0208	0.0129	15.84%	0.7559	0.7238	0.0223	0.0139	17.02%	0.7319	0.7200
	ModernTCN	0.0166	0.0105	13.38%	0.8176	0.7217	0.0179	0.0113	14.34%	0.8096	0.6828
	iTransformer	0.0361	0.0256	32.83%	0.1764	0.4985	0.0383	0.0272	34.74%	0.1242	0.4843
	S2-STI	0.0205	0.0159	19.25%	0.7511	0.7089	0.0211	0.0160	20.25%	0.7425	0.6819
	LLM4HRS	**0.0159**	**0.0100**	**12.50%**	**0.8588**	**0.7402**	**0.0160**	**0.0102**	**12.54%**	**0.8576**	**0.7397**
	Improvement	4.22%↑	4.76%↑	6.58%↑	4.80%↑	2.22%↑	10.62%↑	9.74%↑	12.55%↑	5.60%↑	2.66%↑
	Model	0.7 RMSE	0.7 MAE	0.7 MAPE	0.7 R^2	0.7 SSIM	0.9 RMSE	0.9 MAE	0.9 MAPE	0.9 R^2	0.9 SSIM
	DINEOF	0.0630	0.0520	65.00%	-1.3281	0.3304	0.0807	0.0705	88.09%	-2.8323	0.1413
	Transformer	0.0211	0.0141	18.02%	0.7287	0.6315	0.0250	0.01807	23.03%	0.6282	0.5516
	GURD	0.0239	0.0156	19.94%	0.6537	0.6190	0.0288	0.0201	25.62%	0.4965	0.4906
	DINCAE	0.0247	0.0193	23.48%	0.6352	0.6401	0.0317	0.0266	34.51%	0.4277	0.5735
	Crossformer	0.0255	0.0173	22.08%	0.6163	0.5819	0.0314	0.0223	28.38%	0.4103	0.4873
	TimesNet	0.0387	0.0321	40.94%	0.3237	0.5479	0.0419	0.0344	43.79%	0.2124	0.4670
	SAITS	0.0207	0.0134	17.07%	0.7420	0.6290	0.0235	0.0166	21.12%	0.6375	0.6225
	OFA	0.0241	0.0153	18.73%	0.6850	0.7217	0.0316	0.0213	26.04%	0.4978	0.6856
	ModernTCN	0.0208	0.0150	19.02%	0.7445	0.6148	0.0390	0.0340	43.00%	0.2477	0.5886
	iTransformer	0.0432	0.0308	39.24%	-0.1492	0.4447	0.0445	0.0325	40.39%	-0.1946	0.4780
	S2-STI	0.0217	0.0170	20.82%	0.7376	0.6972	0.0229	0.018	22.05%	0.7214	0.6855
	LLM4HRS	**0.0171**	**0.0111**	**13.56%**	**0.8493**	**0.7327**	**0.0188**	**0.0123**	**14.97%**	**0.8102**	**0.7179**
	Improvement	17.39%↑	17.16%↑	20.56%↑	12.34%↑	1.50%↑	20.00%↑	25.90%↑	29.12%↑	21.32%↑	44.99%↑
	Model	0.2 RMSE	0.2 MAE	0.2 MAPE	0.2 R^2	0.2 SSIM	0.5 RMSE	0.5 MAE	0.5 MAPE	0.5 R^2	0.5 SSIM
YRE-CHA	DINEOF	2.2477	1.3604	30.09%	0.7024	0.6542	3.4214	2.1500	47.59%	0.3111	0.5105
	Transformer	2.1137	1.2988	28.74%	0.7423	0.6945	2.1828	1.3707	30.53%	0.7268	0.6813
	GRUD	2.6181	1.7300	38.28%	0.6050	0.5535	2.7559	1.9160	42.68%	0.5554	0.5469
	DINCAE	2.0696	1.0837	26.56%	0.7927	0.7252	2.7424	1.5149	34.11%	0.7119	0.5658
	Crossformer	2.5089	1.6193	35.83%	0.6343	0.6121	2.5126	1.6243	36.14%	0.6282	0.6064
	TimesNet	2.5452	1.7422	38.55%	0.6245	0.6131	2.5823	1.7959	40.00%	0.6092	0.5885
	SAITS	2.2214	1.3932	30.83%	0.7217	0.6771	2.3622	1.5400	34.30%	0.6888	0.6492
	OFA	1.9169	1.2110	27.16%	0.7895	**0.7320**	1.9422	1.2490	27.80%	0.7768	0.7161
	ModernTCN	1.9415	1.2197	26.54%	0.7819	0.7256	2.0836	1.5004	28.27%	0.7760	0.7010
	iTransformer	2.7691	1.7619	38.99%	0.5683	0.5453	2.9725	1.8862	42.01%	0.5000	0.5264
	S2-STI	2.4857	1.6260	32.83%	0.6528	0.6334	2.5091	1.6461	35.92%	0.6311	0.6198
	LLM4HRS	**1.9053**	**1.2017**	**26.90%**	**0.7906**	0.7268	**1.9094**	**1.2235**	**27.04%**	**0.7883**	**0.7249**
	Improvement	0.61%↑	0.77%↑	0.96%↑	0.14%↑	-0.72%↓	1.69%↑	2.04%↑	2.73%↑	1.46%↑	1.21%↑
	Model	0.7 RMSE	0.7 MAE	0.7 MAPE	0.7 R^2	0.7 SSIM	0.9 RMSE	0.9 MAE	0.9 MAPE	0.9 R^2	0.9 SSIM
	DINEOF	4.4116	2.9320	64.87%	-0.1443	0.3990	5.5287	3.9538	87.34%	-0.7920	0.2000
	Transformer	2.3331	1.4881	33.13%	0.6815	0.6339	2.8000	1.9024	42.32%	0.5476	0.4932
	GRUD	3.0595	2.1040	46.85%	0.4525	0.4577	3.3174	2.3241	51.70%	0.3594	0.4218
	DINCAE	3.2940	1.9911	41.02%	0.5848	0.6826	3.5716	2.2052	43.38%	0.5134	0.5755
	Crossformer	2.5330	1.6235	36.15%	0.6232	0.6019	2.5645	1.6558	36.84%	0.6208	0.6015
	TimesNet	2.6640	1.8871	42.02%	0.5866	0.5552	2.8017	1.9967	44.42%	0.5397	0.4931
	SAITS	2.4265	1.5503	34.52%	0.6584	0.6114	2.5753	1.7316	38.52%	0.6149	0.5840
	OFA	2.1267	1.3539	30.07%	0.7382	0.6871	2.3202	1.5353	33.18%	0.6929	0.6318
	ModernTCN	2.1672	1.5004	33.31%	0.7411	0.6853	2.5797	1.8361	40.73%	0.6290	0.5900
	iTransformer	3.1570	2.0203	45.12%	0.4177	0.4662	3.6258	2.3161	51.52%	0.2392	0.3461
	S2-STI	2.5428	1.6606	36.28%	0.6205	0.5927	2.6832	1.8462	42.50%	0.6002	0.5716
	LLM4HRS	**2.0274**	**1.2930**	**28.72%**	**0.7652**	**0.6913**	**2.2624**	**1.4628**	**31.68%**	**0.7020**	**0.6359**
	Improvement	4.67%↑	4.50%↑	4.49%↑	3.15%↑	6.08%↑	2.49%↑	4.72%↑	4.52%↑	1.30%↑	0.65%↑

1.8.1 and trained using the Adam optimizer [55] on an NVIDIA 4090 GPU. The batch size is set to 16, and the window size is set to 90, meaning that each training sample consists of 90 consecutive time steps. The epoch is fixed to 100, and an early stopping strategy is employed during training to avoid overfitting. The dimension of GPT2 is 768. The mean absolute error (MAE) is used as the loss function. We repeat each experiment five times and select the best parameters for all imputation models through parameter traversal. The initial learning rate is set to 0.001 and will halve after every epoch.

2) Evaluation Metrics: Five evaluation metrics, i.e., the root mean square error (RMSE), MAE, the mean absolute percentage error (MAPE), the coefficient of determination (R^2), and the structural similarity index measure (SSIM), are utilized to evaluate the performance of imputation models, i.e.,

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{t=1}^{\mathrm{T}} \left(\left(X_t - \widehat{X}_t \right) \left(M_t - \overline{M}_t \right) \right)^2}{\sum_{t=1}^{\mathrm{T}} M_t - \overline{M}_t}} \tag{16}$$

$$\mathrm{MAE} = \frac{\sum_{t=1}^{\mathrm{T}} \left| \left(X_t - \widehat{X}_t \right) \left(M_t - \overline{M}_t \right) \right|}{\sum_{t=1}^{\mathrm{T}} M_t - \overline{M}_t} \tag{17}$$

$$\mathrm{MAPE} = \frac{\sum_{t=1}^{\mathrm{T}} \left| \left(X_t - \widehat{X}_t \right) \left(M_t - \overline{M}_t \right) \right|}{\sum_{t=1}^{\mathrm{T}} \left| X_t \left(M_t - \overline{M}_t \right) \right|} \tag{18}$$

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TABLE II

RESULTS OF LLM4HRS AND BASELINE DATA IMPUTATION MODELS ON THE TWO SST DATASETS, I.E., EAST-SST AND YRE-SST, WHILE VARYING THE MASKING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND IMPROVEMENT REPRESENTS THE IMPROVEMENT RATIO OF LLM4HRS AGAINST THE SECOND-BEST RESULTS

Dataset	Model	0.2 RMSE	0.2 MAE	0.2 MAPE	0.2 R^2	0.2 SSIM	0.5 RMSE	0.5 MAE	0.5 MAPE	0.5 R^2	0.5 SSIM
EAST-SST	DINEOF	11.5572	9.6944	35.46%	-20.2229	0.1472	16.4947	15.5320	56.80%	-42.1508	0.0692
	Transformer	0.5274	0.4122	1.66%	0.9554	0.6256	0.6123	0.4760	1.68%	0.9283	0.6053
	GRUD	1.2198	0.8077	2.94%	0.7600	0.5206	1.3451	0.9344	3.40%	0.7082	0.5547
	DINCAE	1.4474	1.1352	4.30%	0.6971	0.5981	1.5133	1.1866	4.49%	0.6670	0.3980
	Crossformer	1.2938	1.0211	3.72%	0.7430	0.3058	1.5325	1.2202	4.42%	0.6180	0.3224
	TimesNet	1.5290	1.0646	4.01%	0.6586	0.4889	1.5641	1.0921	4.12%	0.6160	0.4789
	SAITS	0.6036	0.4717	1.72%	0.9411	0.6241	0.7341	0.5842	2.13%	0.9166	0.6046
	OFA	1.3196	1.0202	3.74%	0.7437	0.6109	1.4321	1.1116	4.08%	0.7148	0.5988
	ModernTCN	1.1324	0.8773	3.20%	0.7889	0.4627	1.1987	0.9685	3.53%	0.7385	0.5893
	iTransformer	2.2803	1.8068	6.58%	0.1557	0.4049	2.3543	1.8528	6.74%	0.1091	0.3487
	LLM4HRS	**0.5129**	**0.3966**	**1.46%**	**0.9631**	**0.6510**	**0.5159**	**0.4032**	**1.48%**	**0.9603**	**0.6212**
	Improvement	2.75%↑	3.79%↑	12.05%↑	0.80%↑	3.90%↑	15.74%↑	15.29%↑	11.91%↑	3.33%↑	2.56%↑
	Model	0.7 RMSE	0.7 MAE	0.7 MAPE	0.7 R^2	0.7 SSIM	0.9 RMSE	0.9 MAE	0.9 MAPE	0.9 R^2	0.9 SSIM
	DINEOF	20.7904	20.2649	74.11%	-67.5687	0.0339	25.2078	25.0031	91.44%	-99.6772	0.0114
	Transformer	0.7257	0.5752	2.09%	0.9150	0.5828	1.0098	0.7849	2.86%	0.8358	0.4716
	GRUD	1.5441	1.1217	4.08%	0.6153	0.4903	1.9969	1.5570	5.67%	0.4576	0.3776
	DINCAE	1.9333	1.6186	6.03%	0.1940	0.3708	2.3537	1.9799	7.37%	0.099	0.2723
	Crossformer	2.0030	2.0518	7.47%	0.2147	0.1962	2.9773	2.5594	9.31%	-0.2918	0.1842
	TimesNet	1.7101	1.2092	4.56%	0.5231	0.4267	1.7566	1.2327	4.65%	0.4549	0.2888
	SAITS	1.4861	1.1983	4.36%	0.6653	0.5535	2.4854	2.1049	7.66%	0.2843	0.2769
	OFA	1.4851	1.1654	4.13%	0.6674	0.5563	1.5139	1.2483	4.23%	0.6547	0.5416
	ModernTCN	1.2904	1.0248	3.73%	0.7094	0.5589	2.1907	1.8081	6.59%	0.3654	0.3879
	iTransformer	2.5971	2.0390	7.42%	-0.0778	0.2687	2.6954	2.1661	7.88%	-0.1809	0.2352
	LLM4HRS	**0.5631**	**0.4380**	**1.61%**	0.9539	0.6187	**0.6787**	**0.5044**	**1.93%**	0.9285	0.6007
	Improvement	22.41%↑	23.85%↑	22.97%↑	4.08%↑	5.80%↑	32.79%↑	35.74%↑	32.52%↑	9.98%↑	21.49%↑
	Model	0.2 RMSE	0.2 MAE	0.2 MAPE	0.2 R^2	0.2 SSIM	0.5 RMSE	0.5 MAE	0.5 MAPE	0.5 R^2	0.5 SSIM
YRE-SST	DINEOF	8.7937	6.9856	31.92%	-1.1817	0.5259	12.8392	11.6761	53.35%	-3.6333	0.2812
	Transformer	1.0147	0.7570	3.42%	0.9508	0.8257	1.5193	1.1986	5.40%	0.9313	0.8230
	GRUD	2.4369	1.5478	6.99%	0.8308	0.7643	2.8596	1.9830	8.94%	0.7658	0.7216
	DINCAE	1.3066	0.9876	4.89%	0.8950	0.7969	1.9355	1.4410	7.45%	0.8406	0.7308
	Crossformer	2.9106	2.3721	10.71%	0.7674	0.6492	3.0791	2.5977	11.70%	0.7078	0.6265
	TimesNet	3.4517	2.6105	11.79%	0.6617	0.7443	3.5087	2.7400	12.35%	0.6493	0.7264
	SAITS	1.4848	1.1554	5.22%	0.9361	0.8111	1.6284	1.2608	5.68%	0.9084	0.8070
	OFA	2.8016	2.1997	10.27%	0.7993	0.7889	3.1157	2.3522	10.95%	0.7435	0.7553
	ModernTCN	2.2435	1.7839	8.07%	0.8765	0.7920	2.3784	1.9046	8.61%	0.8646	0.7744
	iTransformer	3.6879	2.9313	13.24%	0.4340	0.5390	4.3027	3.4524	15.56%	0.3478	0.5092
	LLM4HRS	**0.8339**	**0.6146**	**2.87%**	**0.9813**	**0.8509**	**0.8928**	**0.6899**	**3.21%**	**0.9781**	**0.8478**
	Improvement	17.82%↑	18.81%↑	16.08%↑	3.11%↑	2.96%↑	41.24%↑	42.44%↑	40.56%↑	4.79%↑	2.93%↑
	Model	0.7 RMSE	0.7 MAE	0.7 MAPE	0.7 R^2	0.7 SSIM	0.9 RMSE	0.9 MAE	0.9 MAPE	0.9 R^2	0.9 SSIM
	DINEOF	16.6602	15.7222	71.84%	-6.7992	0.1282	20.6944	19.8917	90.90%	-11.0317	0.0338
	Transformer	2.1836	1.7141	7.72%	0.8541	0.7414	3.2370	2.5991	11.71%	0.7137	0.5973
	GRUD	3.0726	2.1524	9.70%	0.7298	0.6906	5.0846	3.9855	17.97%	0.2626	0.4820
	DINCAE	2.5152	1.9488	9.97%	0.7800	0.6809	3.3057	2.6623	13.84%	0.6659	0.5804
	Crossformer	3.3506	2.7354	12.34%	0.6869	0.6115	4.6316	4.0000	18.03%	0.2688	0.3936
	TimesNet	3.6434	2.8547	12.86%	0.6127	0.6132	6.0596	5.1451	23.19%	0.1080	0.3216
	SAITS	2.1373	1.6423	7.40%	0.8718	0.7627	2.9158	2.2755	10.26%	0.7591	0.6777
	OFA	3.4413	2.5887	12.05%	0.6921	0.7494	3.6373	2.7578	12.84%	0.6477	0.7414
	ModernTCN	2.4993	2.0186	9.12%	0.7923	0.7012	5.6101	4.6615	21.06%	0.1918	0.4395
	iTransformer	4.5781	3.7354	16.82%	0.3036	0.5025	4.9246	3.9607	17.85%	0.2686	0.4623
	LLM4HRS	**1.0128**	**0.7909**	**3.68%**	**0.9703**	**0.8395**	**1.3480**	**1.0326**	**4.81%**	**0.9517**	**0.8346**
	Improvement	52.61%↑	51.84%↑	50.27%↑	10.15%↑	9.15%↑	53.77%↑	54.62%↑	53.12%↑	20.24%↑	18.80%↑

$$R^2 = 1 - \frac{\sum_{t=1}^{\mathrm{T}} \left(\left(X_t - \widehat{X}_t \right) \left(M_t - \overline{M}_t \right) \right)^2}{\sum_{t=1}^{\mathrm{T}} \left(\left(X_t - X_{\mathrm{mean}} \right) \left(M_t - \overline{M}_t \right) \right)^2} \tag{19}$$

$$\mathrm{SSIM} = \frac{\left(2\mu_X \mu_{\widehat{X}} + C_1 \right) \left(2\sigma_{X\widehat{X}} + C_2 \right)}{\left(\mu_X^2 + \mu_{\widehat{X}}^2 + C_1 \right) \left(\sigma_X^2 + \sigma_{\widehat{X}}^2 + C_2 \right)} \tag{20}$$

where X_t and $\widehat{X}_t$ are the true value and generated values at time step t for all locations; M_t and $\overline{M}_t$ are the corresponding masks for the true value X_t and the residual value $\overline{X}_t$ at time step t, respectively; the residual values $\overline{X}_t$ are obtained by randomly masking a specified proportion of the true values X_t; μ_X and $\mu_{\widehat{X}}$ are the average values of X and $\widehat{X}$, respectively; σ_X^2 and $\sigma_{\widehat{X}}^2$ are the variances of X and $\widehat{X}$, respectively; $\sigma_{X\widehat{X}}$ is the covariance between X and $\widehat{X}$; and C_1 and C_2 are small constants used to maintain stability. Note that the missing values in the datasets are not evaluated due to the absence of ground truth.

3) Parameters' Setting: The optimal number of EOFs retained in DINEOF is 15 for YRE-CHA, seven for both EAST-CHA and YRE-SST, and ten for EAST-SST. In GRUD, the optimal hidden dimension is 768 for YRE-CHA and 2048 for the other datasets. The parameters for DINCAE follow those reported in the original paper. In TimesNet, the top-k amplitude values are set to one for EAST-CHA, four for YRE-SST, and five for both YRE-CHA and EAST-SST. The optimal hidden dimensions in ModerTCN are [32, 32, 32, 32] for all datasets. Other baseline models, including Transformer, Crossformer, SAITS, OFA, and iTransformer,

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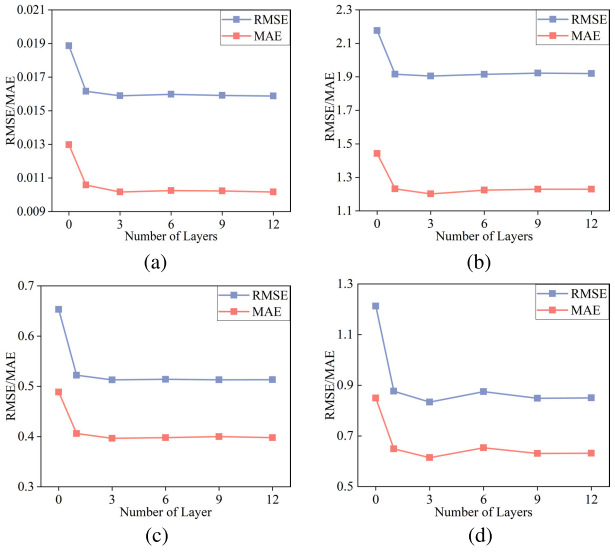


Fig. 7. RMSE and MAE of LLM4HRS on all datasets while varying the number of model layers, where the data masking rate is set to 0.2. (a) EAST-CHA. (b) YRE-CHA. (c) EAST-SST. (d) YRE-SST.

are attention-based models, and the number of layers is their primary parameter. In Transformer and iTransformer, the layer configuration is set to six for EAST-SST and two for the other datasets. In Crossformer, SAITS, and OFA, the numbers of layers are set to 2, 4, and 3, respectively, for all datasets. Note that the above settings are based on the parameter optimization experiments, which are performed for all baseline models with a masking rate of 0.5.

4) Training and Testing Datasets: For all experiments, each dataset is divided chronologically into training, validation, and testing datasets. Concretely, we use 80% of each dataset for training, and the remaining 20% is equally divided into validation and testing datasets.

5) Number of Model Layers: For data imputation, the number of layers in the GPT2 model is essential. Too few layers can lead to underfitting, while too many layers may result in overfitting. To better balance imputation performance and computational efficiency, we test the performance of LLM4HRS while using different numbers of model layers. Fig. 7 shows the RMSE and MAE of LLM4HRS while varying the number of layers, where the data masking rate is set to 0.2. According to the results, we set the number of layers to 3 for all datasets.

C. Experimental Results and Analysis

Tables I and II show the data imputation performance metrics, including RMSE, MAE, MAPE, R^2 and SSIM, of LLM4HRS model and baseline models for sparse Chl-a data and SST data in the two study areas while varying the data masking rate. The masking rate refers to the proportion of valid observations randomly masked in the original data matrix $\mathcal{X}$. The results indicate that LLM4HRS consistently outperforms all baseline models on all datasets, especially when dealing with datasets with high masking rates. Specifically, under conditions of a relatively high masking rate, the LLM4HRS model achieves improvements of up to 20.00% and 25.90% for Chl-a and 53.77% and 54.62% for SST in terms of RMSE and MAE, respectively.

According to Tables I and II, the statistical model DINEOF performs poorly when applied to highly sparse remote sensing data, as it struggles to effectively capture complex spatial features. iTransformer performs the worst among attention-based models because it is a spatial-centric attention model that cannot effectively capture the temporal dependencies. TimesNet performs poorly, which is attributed to its inaccurate extraction of periodic information from highly sparse remote sensing data. ModernTCN performs well at low masking rates, especially on the Chl-a dataset. However, its performance is inferior to TimesNet at high masking rates due to the limited information propagation ability that prevents effective capture of temporal dependencies. Crossformer performs moderately since its full connection mechanism for capturing spatial dependencies tends to introduce noise. The performance of GRUD, which captures only temporal dependencies, is similar to that of DINCAE, which focuses solely on spatial dependencies. This similarity indicates that relying exclusively on either temporal or spatial dependencies is insufficient for accurately imputing highly sparse remote sensing data. When applied to highly sparse remote sensing data, the traditional Transformer performs well under low masking rates. However, as it is not specifically designed for data imputation tasks, its performance declines under high masking rates. SAITS introduces the diagonally masked self-attention for data imputation and shows good performance. Similarly, OFA also performs well at high masking rates since it utilizes the high representation learning ability of GPT2. Notably, OFA achieves the second-best imputation performance at a high masking rate of 0.9 on the EAST-CHA, YRE-CHA, and YRE-SST datasets, further indicating the potential of LLMs for imputing highly sparse remote sensing data.

According to Tables I and II, with the increase in the masking rate, the performance of all the imputation models decreases due to the reduction of available remote sensing observation data for model training and data imputation. It is obvious that as the masking rate increases, LLM4HRS demonstrates the smallest performance decline across most datasets, further validating its high imputation accuracy for highly sparse remote sensing data.

D. Ablation Studies

To validate the effectiveness of key individual components of the model, we introduce different variants of LLM4HRS as below.

1) *w/o T:* Remove the LLM-based bidirectional temporal representation learning.
2) *w/o S:* Remove the LLM-based spatial representation learning.
3) *w/o F:* Remove the spatiotemporal representation fusion, and retain only the LLM-based spatial representation learning to confirm the importance of spatiotemporal representation fusion.
4) *w/o B:* Remove the LLM-based bidirectional temporal representation learning, but retain the LLM-based

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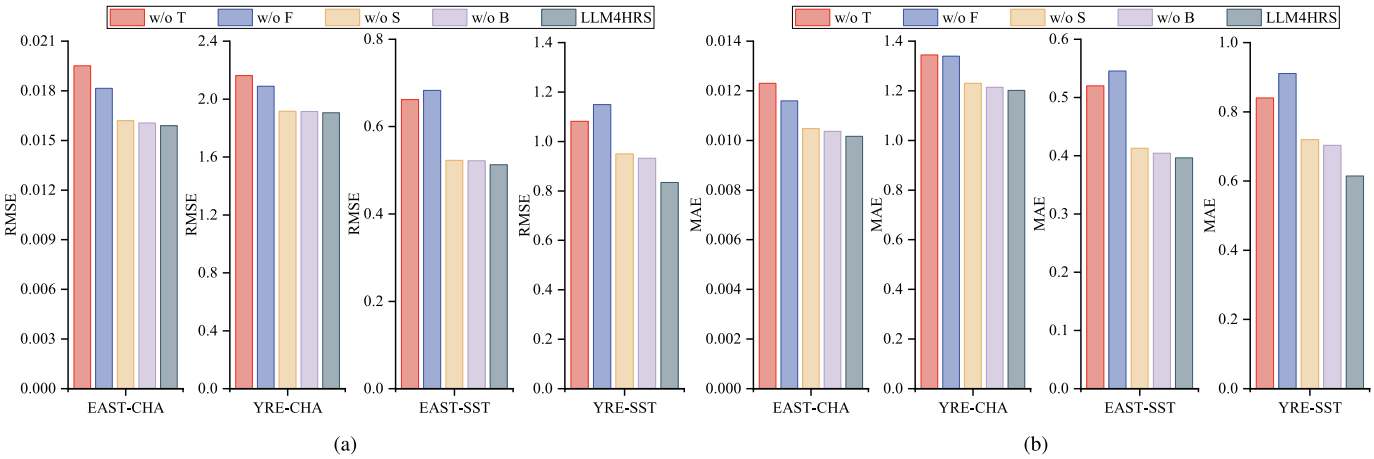


Fig. 8. (a) RMSE and (b) MAE of LLM4HRS and its variants after removing certain components.

TABLE III
RMSE AND MAE OF LLM4HRS AND ITS VARIANTS, I.E., LLM4HRS(W/O S) AND LLM4HRS(W/O T), UNDER A MASKING RATE OF 0.9

Model	EAST-CHA		YRE-CHA		EAST-SST		YRE-SST	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
LLM4HRS(w/o S)	0.0205	0.0134	2.3490	1.5023	0.7430	0.5765	1.5079	1.1588
LLM4HRS	0.0188	0.0123	2.2624	1.4628	0.6787	0.5044	1.3480	1.0326
Improvement	8.29%↑	8.21%↑	3.69%↑	2.63%↑	8.65%↑	12.51%↑	10.60↑	10.89%↑
LLM4HRS(w/o T)	0.0221	0.0140	2.4110	1.5809	0.8189	0.6313	1.5693	1.1859
LLM4HRS	0.0188	0.0123	2.2624	1.4628	0.6787	0.5044	1.3480	1.0326
Improvement	14.93%↑	12.14%↑	6.16%↑	7.47%↑	17.12%↑	20.10%↑	14.10↑	12.93%↑

forward temporal representation learning to assess the necessity of bidirectional temporal representation learning.

Fig. 8 shows the results of these variants and the original LLM4HRS, where the data masking rate is set to 0.2. According to the results, all the variants perform worse than LLM4HRS, which indicates the effectiveness and necessity of the individual components. Specifically, in the case of w/o T, the LLM4HRS model performs almost the worst on all datasets, indicating that LLM-based bidirectional temporal representation learning effectively captures the temporal patterns in remote sensing data. The performance of w/o F is almost the second worst, further demonstrating the effectiveness of LLM-based bidirectional temporal representation learning for data imputation. The importance of spatial representation learning for data imputation is also evident according to the performance of w/o S. Finally, the results of w/o B confirm the importance of backward temporal representation for data imputation. In sum, the LLM-based bidirectional temporal representation learning module and the LLM-based spatial representation learning module are essential for the LLM4HRS model to achieve good data imputation performance for sparse remote sensing data.

We further quantify the contributions of the spatial representation learning module via the variant LLM4HRS (w/o S) and the bidirectional temporal representation learning module via LLM4HRS (w/o T). As shown in Table III, LLM4HRS (w/o S) exhibits significant performance drops, which underscores the importance of the spatial representation learning module in enhancing the overall model performance of LLM4HRS. When the bidirectional temporal module is removed, i.e., LLM4HRS (w/o T), the RMSE and MAE of LLM4HRS decrease by 6.16%–17.12% and 7.47%–20.10%, respectively. This indicates that the bidirectional temporal representation learning module is one critical component for improving LLM4HRS's performance.

VI. DISCUSSION

LLMs are trained on vast datasets that contain rich sequential features, enabling LLMs to demonstrate significant abilities in capturing sequential patterns. Through fine-tuning or prompt engineering, the robust sequential modeling capabilities of LLMs can be fully utilized to capture complex patterns in time series data. To address the challenge of limited effective information in highly sparse remote sensing data, we aim to fully leverage the advantages of LLMs in capturing sequential patterns by designing temporal and spatial representation learning modules based on LLMs. Specifically, we propose a bidirectional temporal representation learning module based on LLMs to better capture the temporal information in highly sparse remote sensing data. Meanwhile, to capture the spatial dependencies in remote sensing data, we also design a spatial representation learning module based on LLMs. To clearly demonstrate the advantages of LLM4HRS, we carry out the discussion on its excellent ability in learning the spatiotemporal representation during the data imputation process on the YRE-CHA and YRE-SST datasets, respectively. We also validated the effectiveness of LLM4HRS by comparing it with in situ data. For simplicity, we primarily discuss the differences between LLM4HRS and

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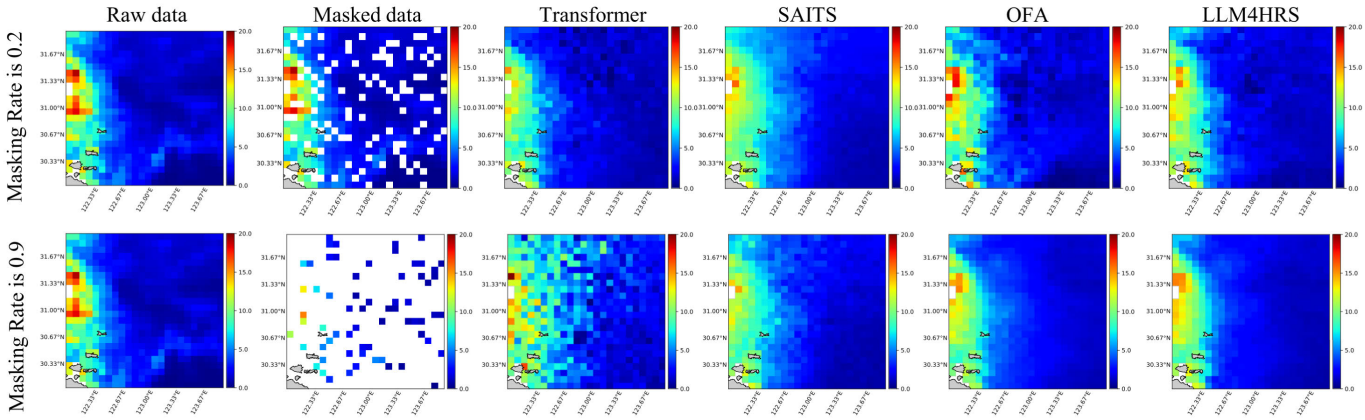


Fig. 9. Raw Chl-a data sample (September 18, 2023) in the YRE-CHA dataset, along with the corresponding masked data at masking rates of 0.2 and 0.9, and the imputed data using Transformer, SAITS, OFA, and LLM4HRS.

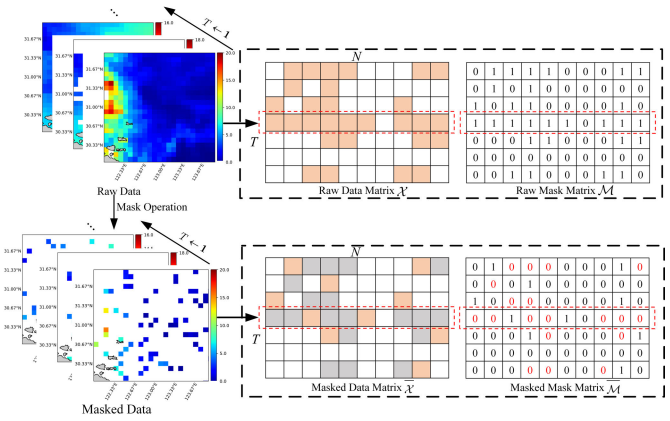


Fig. 10. Visualization of the changes in the data matrix and the mask matrix before and after the masking operation.

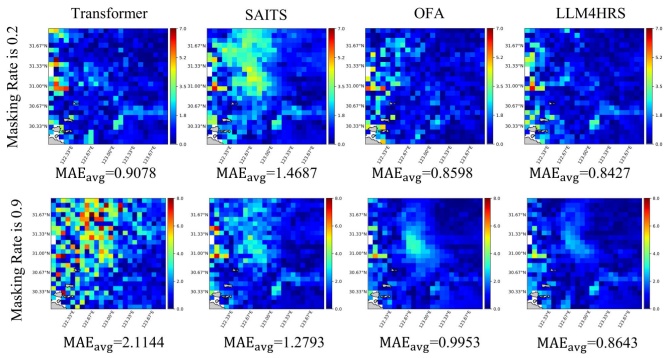


Fig. 11. MAE of Transformer, SAITS, OFA, and LLM4HRS for imputing masked Chl-a on September 18, 2023.

the three most competitive baseline models, i.e., Transformer, SAITS, and OFA. In addition, we also discuss the applicability of LLM4HRS and its computation efficiency.

A. Superiority of LLM4HRSfor Chl-a Data

Fig. 9 illustrates the raw Chl-a data sample in the YRE-CHA dataset, along with the corresponding masked data at masking rates of 0.2 and 0.9, and the imputed data using Transformer, SAITS, OFA, and LLM4HRS. We chose the data sample with almost no missing values because it has more true values to evaluate the accuracy of the filled values by data imputation models. The results show that under low mask data conditions, the generated data align with the original data, with the LLM4HRS model demonstrating particularly superior performance. Specifically, the LLM4HRS model exhibits lower trend errors under these conditions. This indicates that a lower masking rate provides more effective information, allowing the model to learn more effectively and reduce trend errors. In addition, to clearly illustrate the changes in the raw data matrix and the mask matrix before and after the masking operation, we provide the corresponding schematic, as shown in Fig. 10. It can be observed that the masking operation applies solely to the observed values in the original data, leaving the locations of missing data unprocessed. The masked data refer to the remaining data after randomly masking a certain proportion of the observed values in the original dataset.

Fig. 11 also visualizes the MAE of Transformer, SAITS, OFA, and LLM4HRS on the selected data sample. The visualizations indicate that the LLM4HRS model has a clear advantage over Transformer, SAITS, and OFA, further validating the effectiveness of the bidirectional temporal and spatial representation learning modules. Overall, the LLM4HRS model demonstrates a relatively low MAE, especially in regions with highly complex spatiotemporal dependencies, such as red tide-prone areas. This further confirms that the LLM4HRS model can adequately learn the spatiotemporal representation of Chl-a data even with extremely limited information.

B. Superiority of LLM4HRSfor SST Data

Figs. 12 and 13 present the visualizations of the generated SST and MAE of the LLM4HRS model and several competing baseline models for specific data samples under a random masking rate of 0.2 and 0.9, respectively. According to the results, our model also achieves the best overall performance. In addition, we quantitatively analyzed the relative variability of SST and Chl-a using the dimensionless coefficient of

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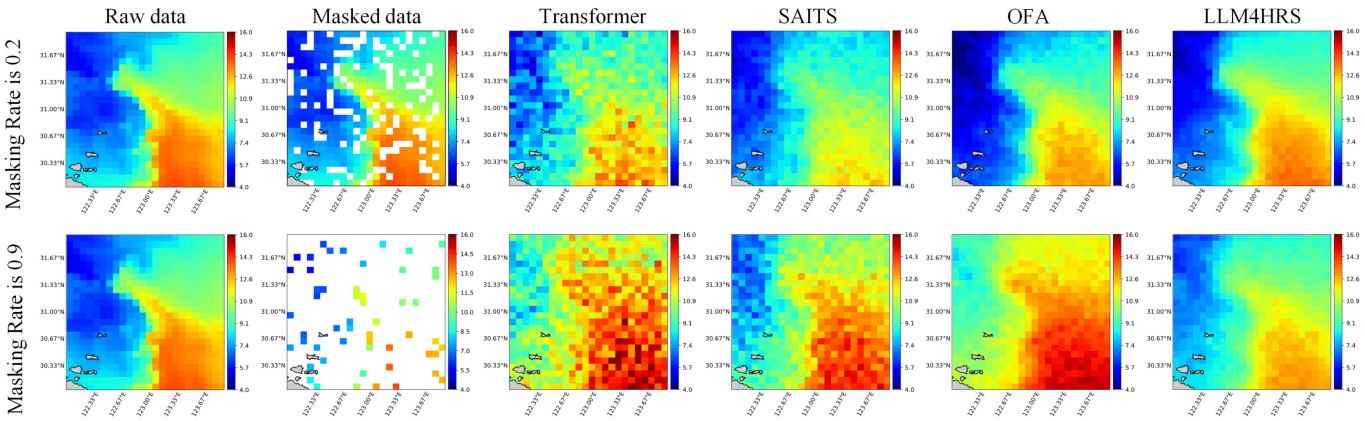


Fig. 12. Raw SST data sample (January 31, 2023) in the YRE-SST dataset, along with the corresponding masked data at the masking rate of 0.2 and 0.9, and the imputed data by Transformer, SAITS, OFA, and LLM4HRS.

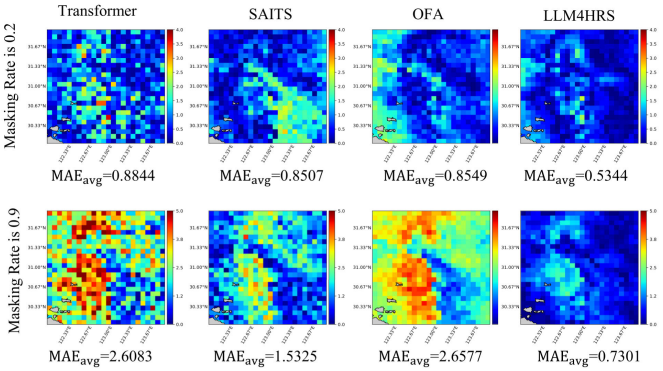
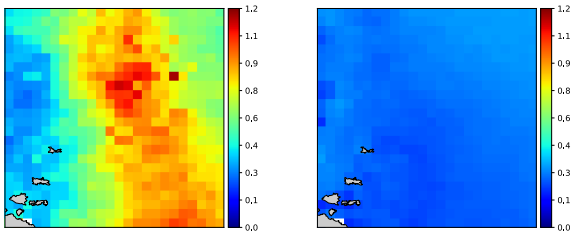


Fig. 13. MAE of Transformer, SAITS, OFA, and LLM4HRS for imputing masked SST on January 31, 2023.



(a) (b)

Fig. 14. CV distributions of Chl-a and SST in the study area of Yangtze River Estuary. (a) YRE-CHA. (b) YRE-SST.

variation (CV), i.e.,

$$\mathrm{CV} = \frac{\mu}{\sigma} \tag{21}$$

where μ is the mean and σ is the standard deviation. A smaller CV indicates lower variability and more stable trends.

Fig. 14 illustrates the CV distributions of SST and Chl-a data in the YRE study area. It is obvious that SST data consistently exhibit low and smooth CV values, with a range of 0–0.3. In contrast, Chl-a data display a high CV range of 0.4–1.2, characterized by pronounced localized variability. The smooth spatial distributions and low variability of SST data facilitate the capture of temporal trends, thus enabling LLM4HRS model to leverage its long-sequence learning capabilities effectively to capture long-term trends of SST. On the other hand, Chl-a data are heavily influenced by local marine ecosystem changes, e.g., short-term phytoplankton fluctuations. This results in greater variability and uncertainty, thus increasing the difficulty for LLM4HRS to learn the long-term trends of Chl-a data.

C. Superiority of LLM4HRS With Cloud Masking

To evaluate the effectiveness of LLM4HRS when applied to actual remote sensing image completion, we adopt the block missing patterns from general time series data imputation [56] and design a hybrid masking strategy that combines random discrete masking and continuous region masking to simulate cloud masks. Specifically, for each location, we randomly discard 5% of the available data. In addition, to simulate the continuous data masking due to cloud cover, we set the cloud occurrence probability to 5% and uniformly sample a continuous region, where the size of the sampled region is also randomly selected. Table IV presents a performance comparison of LLM4HRS and multiple competitive data imputation models under the cloud mask mode in the YRE study area. As shown in Table IV, LLM4HRS demonstrates a clear advantage in the cloud mask mode. Furthermore, Figs. 15 and 16 provide visualizations of the data reconstruction, with the selected samples for visualization consistent with those using the random mask, respectively. These visualizations further validate the superior performance of LLM4HRS in data reconstruction under the cloud mask mode.

D. Validation of LLM4HRS With In Situ Data

We utilize the in situ data from the National Marine Data Center.[3] We specifically select the daily measurements from the Sheng Shan Station (locations: 122.80°E, 30.80°N) during the period from January 2022 to December 2023, retaining 31 valid daily samples that coincided temporally with remote sensing data. Table V gives the relationship between reconstructed values and corresponding in situ data. Specifically, the RMSE (1.2259) of LLM4HRS is reduced by

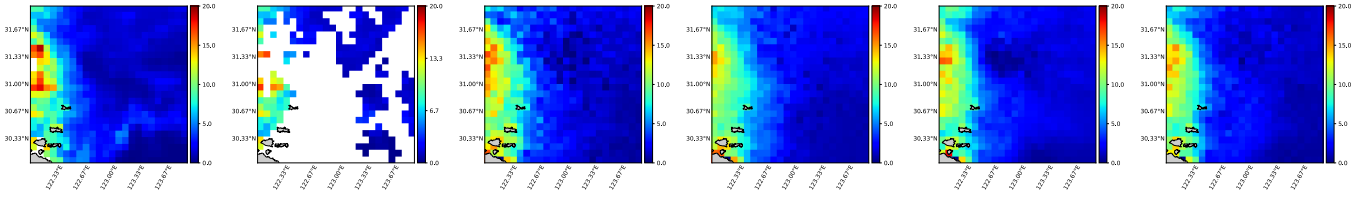
[3]https://mds.nmdis.org.cn/

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TABLE IV

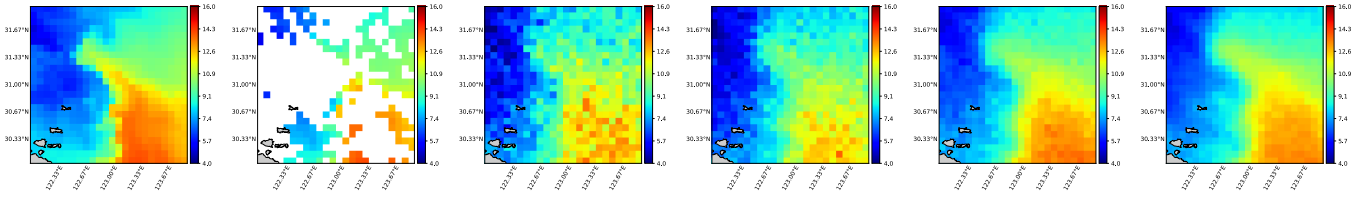
RESULTS OF LLM4HRS AND THREE COMPETITIVE DATA IMPUTATION MODELS WITH CLOUD MASK ON THE YRE STUDY AREA, I.E., YRE-CHA AND YRE-SST, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND IMPROVEMENT REPRESENTS THE IMPROVEMENT RATIO OF LLM4HRS AGAINST THE SECOND-BEST RESULTS

Model	YRE-CHA					YRE-SST				
	RMSE	MAE	MAPE	R^2	SSIM	RMSE	MAE	MAPE	R^2	SSIM
Transformer	2.3968	1.5533	34.52%	0.6664	0.6051	2.0100	1.5773	7.10%	0.9040	0.6972
SAITS	2.4531	1.5972	35.50%	0.6505	0.6068	1.7608	1.3554	6.12%	0.9110	0.7729
OFA	2.0562	1.3008	28.88%	0.7548	0.6878	1.5293	1.2597	6.06%	0.9393	0.8501
LLM4HRS	**1.9765**	**1.2575**	**27.87%**	**0.7735**	**0.7056**	**0.8790**	**0.6725**	**3.29%**	**0.9825**	**0.8941**
Improvement	3.88%↑	3.33%↑	3.50%↑	2.42%↑	2.52%↑	42.52%↑	46.61%↑	45.71%↑	4.40%↑	4.92%↑



(a) (b) (c) (d) (e) (f)

Fig. 15. Raw Chl-a data sample (September 18, 2023), the corresponding masked data with cloud mask, and the imputed data by Transformer, SAITS, OFA, and LLM4HRS in the YRE-CHA dataset. (a) Raw data. (b) Masked data. (c) Transformer. (d) SAITS. (e) OFA. (f) LLM4HRS.



(a) (b) (c) (d) (e) (f)

Fig. 16. Raw SST data sample (January 31, 2023), the corresponding masked data with cloud mask, and the imputed data by Transformer, SAITS, OFA, and LLM4HRS in the YRE-SST dataset. (a) Raw data. (b) Masked data. (c) Transformer. (d) SAITS. (e) OFA. (f) LLM4HRS.

TABLE V

COMPARISON BETWEEN THE RECONSTRUCTED VALUES AND THE CORRESPONDING IN SITU DATA

Model	RMSE	MAE	R^2	Slope	Intercept
Transformer	2.2737	2.1236	0.8265	0.9870	2.4034
SAITS	1.9892	1.7344	0.8672	1.2214	-3.9972
OFA	2.2209	1.6355	0.8344	0.8829	2.8239
LLM4HRS	**1.2259**	**0.9587**	**0.9496**	**1.0208**	**0.2474**

62.26% compared to the second-best model, and R^2 (0.9496) is improved by 9.50%. Furthermore, as shown in Fig. 17, the comparisons between the LLM4HRS and several competitive data imputation models against in situ data reveal that the LLM4HRS reconstruction results align closely with the ideal line. These results suggest that the LLM4HRS-reconstructed data better reflect both statistical significance and trend consistency with in situ observations.

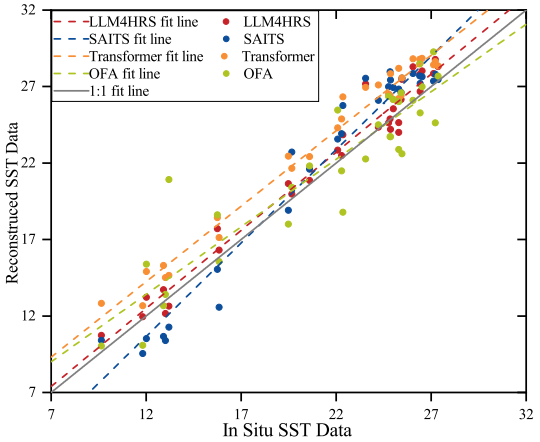


Fig. 17. Comparison of the reconstructed SST data from the four reconstruction models, with in situ SST data, where the dotted lines represent the fit lines for the corresponding scatter points of four models and the gray solid line represents the ideal line.

E. Applicability of LLM4HRS

To evaluate the applicability of LLM4HRS for large datasets, we assessed its performance on a large dataset (referred to as SOUTH-PAR) that covers the area of [110.00°E–120.00°E, 10.00°N–20.00°N]. Table VI compares LLM4HRS with multiple competitive data imputation models, demonstrating that LLM4HRS could scale for large datasets and consistently delivers superior imputation performance. Moreover, for a large ocean area, we could also simply employ a downsampling strategy to reduce the number of grid regions to improve efficiency if the requirement for spatial resolution could be relaxed.

F. Computation Efficiency

Table VII presents the number of trainable parameters (in millions) and the running time of each training epoch (in

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TABLE VI

RESULTS OF LLM4HRSAND THREE COMPETITIVE DATA IMPUTATION MODELS ON THE PAR DATASET, I.E., SOUTH-PAR, WHILE VARYING THE MASKING RATE, WHERE THE **BOLD RESULTS** ARE THE BEST, THE UNDERLINED RESULTS ARE THE SECOND BEST, AND IMPROVEMENT REPRESENTS THE IMPROVEMENT RATIO OF LLM4HRSAGAINST THE SECOND-BEST RESULTS

Model	0.2					0.5				
	RMSE	MAE	MAPE	R^2	SSIM	RMSE	MAE	MAPE	R^2	SSIM
Transformer	10.1781	7.1298	17.56%	0.4454	0.2454	10.3673	7.1424	17.58%	0.4255	0.2517
SAITS	9.4808	6.6135	16.28%	0.5188	0.2843	10.4457	7.3409	18.06%	0.4168	0.2738
OFA	8.1451	5.8895	14.21%	0.6550	0.4161	9.0928	6.6284	15.97%	0.5698	0.3528
LLM4HRS	**6.6675**	**4.5473**	**10.97%**	**0.7688**	**0.5148**	**6.9223**	**4.7308**	**11.40%**	**0.7507**	**0.4709**
Improvement	18.14%↑	22.79%↑	22.80%↑	14.80%↑	19.17%↑	30.10%↑	28.63%↑	28.62%↑	24.10%↑	25.08%↑
Model	0.7					0.9				
	RMSE	MAE	MAPE	R^2	SSIM	RMSE	MAE	MAPE	R^2	SSIM
Transformer	10.6409	7.2168	17.77%	0.3963	0.2473	10.8759	7.6784	18.90%	0.3699	0.2305
SAITS	10.6730	7.7364	19.04%	0.3927	0.2655	11.1847	7.7065	18.97%	0.3336	0.2335
OFA	9.5389	6.9897	16.85%	0.5281	0.3311	11.2137	8.3245	20.08%	0.3489	0.3119
LLM4HRS	**7.4013**	**5.0886**	**12.49%**	**0.7159**	**0.4233**	**8.4143**	**5.9139**	**14.26%**	**0.6334**	**0.3616**
Improvement	22.41%↑	27.20%↑	25.88%↑	26.23%↑	21.78%↑	24.96%↑	28.96%↑	28.98%↑	44.92%↑	13.74%↑

TABLE VII

NUMBER OF TRAINABLE PARAMETERS (IN MILLION) AND THE RUNNING TIME OF EACH TRAINING EPOCH (IN SECONDS) FOR LLM4HRS AND THE BASELINE MODELS ON THE CHA-YRE DATASET

Model	Parameters (M)	Time Cost (seconds/epoch)
GRUD	7.42	1.2
Transformer	6.84	5.4
DINCAE	0.10	10.22
OFA	2.57	7.9
ModernTCN	24.23	19.8
LLM4HRS	4.78	30.1
LLM4HRS (w/o S)	4.16	10.2

seconds) for LLM4HRS and competitive baseline models on the CHA-YRE dataset. Though LLM4HRS is based on LLMs, it has a small number of trainable parameters since most of the parameters in LLMs are frozen. In terms of time cost, the training process of LLM4HRS remains time-intensive due to the bidirectional temporal and spatial representation learning. However, the time cost of LLM4HRS is still comparable with the baseline models.

We also present the variant of LLM4HRS that removes the spatial representation learning module (LLM4HRS w/o S), which significantly reduces the training time and indicates that the majority of LLM4HRS's training time is dedicated to the learning of spatial representations. According to Table III, LLM4HRS (w/o S) exhibits significant drops in performance, which indicates that the spatial representation module is crucial for improving the overall model performance. Although spatial representation learning requires considerable training time, the performance gains justify the additional computational cost. In fact, compared with the baseline models in Table VII, the time cost of the LLM4HRS model is still within a reasonable and acceptable range.

VII. CONCLUSION

In this work, we proposed a novel LLM-based spatiotemporal model termed LLM4HRS for highly sparse remote sensing data imputation. The LLM-based partially frozen strategy fully learns the forward and backward temporal dependencies in remote sensing data. Furthermore, leveraging the LLM-based spatial representation learning, we address the deficiency issue of existing LLM-based models in effectively learning spatial correlations. The extensive experiments on sparse datasets for Chl-a and SST demonstrate that the LLM4HRS model outperforms state-of-the-art models significantly, showcasing the potential of LLM in highly sparse remote sensing data imputation.

In future work, we plan to study collaborative data imputation of multisource ocean data and fully leverage the correlations between ocean factors to further enhance the data imputation accuracy. In addition, incorporating oceanographic knowledge into our LLM4HRS model to enrich the available information for highly sparse remote sensing data imputation is also a promising research problem.

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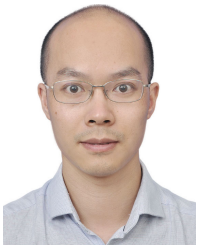
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